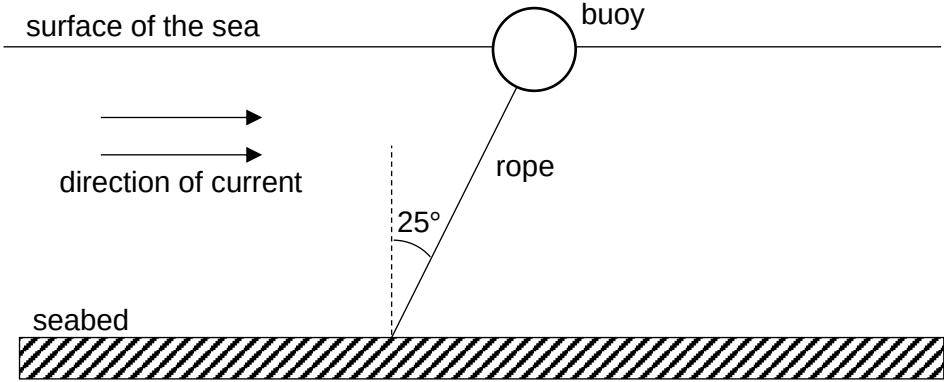
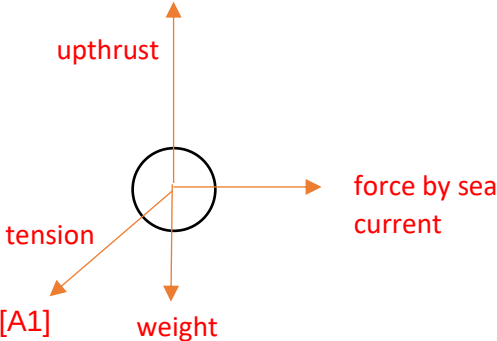


3	(a)	Explain the origin of upthrust.	
			[1]
		Upthrust is due to the <u>pressure difference between upper and lower surfaces/points of the object in the fluid.</u> [B1]	
	(b)	A buoy is attached to a rope which is anchored to the seabed. Currents in the sea cause the buoy to be displaced so that the rope makes an angle of 25° with the vertical, as shown in Fig. 3.1. The buoy is in equilibrium.  Fig. 3.1 The currents in the sea exert a horizontal force of 140 N on the buoy and the weight of the buoy is 130 N.	
	(i)	Sketch the forces acting on the buoy in Fig. 3.2.	
		 Fig. 3.2	
			[2]
		 Correct forces [M1] Correct relative magnitude [A1]	

		(ii)	Show that the upthrust acting on the buoy is 430 N.
			[2]
			Horizontal component of tension in rope = force of sea current $T \sin 25^\circ = 140$ Vertical component of tension in rope + weight of buoy = upthrust on buoy $T \cos 25^\circ + 130 = U$ [B1 expressions for forces] $\tan 25^\circ = \frac{140}{U - 130}$ [M1 manipulation of equations] $U = 430 \text{ N}$ [A0]
		(iii)	Determine the percentage of the volume of the buoy that is submerged in the sea. The density of sea water is 1000 kg m^{-3} and density of the buoy is 230 kg m^{-3} .
			percentage of volume of buoy submerged = % [3]
			Volume of the buoy = $\frac{m}{\rho} = \frac{130/9.81}{230} = 0.0576 \text{ m}^3$ [C1] Upthrust = $V\rho g$ Volume of buoy submerged = $\frac{430}{1000 \times 9.81} = 0.0438 \text{ m}^3$ [C1] Percentage of volume submerged = $\frac{0.0438}{0.0576} \times 100\% = 76\%$ [A1]
			[Total: 8]

4

A rocket is midway between the Earth and Moon as shown in Fig. 4.1.